1 Mathematical Formulation

1.1 Index Sets

- $K = \{1, 2, \ldots, n_K\}$ as the set of all orders
- $K^0 = K \cup \{0\}$
- $C = \{1, 2, \ldots, n_C\}$ as the set of all cars

1.2 Parameters

- R_k: Sales revenue of accepting order k for all $k \in K$
- S_k^{Start}: Pick up station of accepting order k for all $k \in K$
- S_k^{End}: Pick up station of accepting order k for all $k \in K$
- S_c^0: The initial station for the car c for all $c \in C$
- B: Maximum total moving time allowed (in minutes).
- L_c: The designated level of car c. for all $c \in C$
- Req_k: The required car level for order k for all $k \in K$
- T_k^S: The start time for order k for all $k \in K$
- T_k^E: The end time for order k for all $k \in K$
- $T_{j,k}^{Trans}$: The transport time from order j to order k for all $j, k \in K$
- $T_{0,j}^{Trans}$: The transport time from the initial station to order j for all $j \in K$

1.3 Decision Variables

- $x_k \in \{0, 1\}$: 1 if order k is accepted, 0 otherwise.
- $y_{c,k} \in \{0, 1\}$: 1 if car c is assigned to execute order k, 0 otherwise.
- $u_k \in \{0, 1\}$: 1 if a free upgrade is provided for order k, 0 otherwise.
- $z_{c,i,j} \in \{0, 1\}$: 1 if car c executes order j immediately after finishing order i, 0 otherwise.

1.4 Objective Function

The objective is to maximize the total profit, which is the total sales revenue minus the total compensation for rejected orders:

$$\text{Maximize} \quad \sum_{k \in \mathcal{K}} R_k x_k - \sum_{k \in \mathcal{K}} 2R_k(1 - x_k) \tag{1}$$

1.5 Constraints

$$\sum_{c\in C} y_{c,k} = x_k \qquad \forall k \in K \tag{2}$$

$$\sum_{c\in C} L_c y_{c,k} = Req_k x_k + u_k \qquad \forall k \in K \tag{3}$$

$$\sum_{j\in K^0, j\neq k} z_{c,j,k} = y_{c,k} \qquad \forall k \in K \quad \forall c \in C \tag{4}$$

$$\sum_{k\in K, j\neq k} z_{c,j,k} \leq y_{c,j} \qquad \forall j \in K \quad \forall c \in C \tag{5}$$

$$\sum_{j\in K} z_{c0j} \leq 1 \qquad \forall c \in C \tag{6}$$

$$T_k^S - T_j^E + V(1 - z_{c,j,k}) \geq 9 + T_{j,k}^{Trans} \qquad \forall c \in C \quad j \neq k \quad \forall j \in K \quad \forall k \in K \tag{7}$$

$$T_k^S + V(1 - z_{c,0,k}) \geq 1 + T_{0,k}^{trans} \qquad S_c^0 \neq S_k^{Start} \quad \forall c \in C \quad \forall k \in K \tag{8}$$

$$T_k^S + V(1 - z_{c,0,k}) \geq T_{0,k}^{trans} \qquad S_c^0 = S_k^{Start} \quad \forall c \in C \quad \forall k \in K \tag{9}$$

$$\sum_{c\in C} \sum_{j\in K^0, j\neq k} \sum_{k\in K} T_{j,k}^{Trans} z_{c,j,k} \leq B \tag{10}$$